

Dilli Ram Chaudhary

1. Install Python on your system and verify the installation by printing "Hello, World!" in both interactive mode and as a script file.
2. Create variables of different data types (int, float, str, bool) and use the type() function to identify each one.
3. Write a program that takes two numbers as input from the user and prints their sum, difference, product, and division.
4. Store your full name in a string variable and print its length, uppercase version, lowercase version, and replace spaces with underscores.
5. Create a list of 5 favorite movies, print each movie with its index number (starting from 1), then ask the user to add one more movie and display the final list.
6. Write a program that takes a sentence from the user and counts how many times the letter 'e' appears (case-insensitive).

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In [2]: # 2. Create variables of different data types(int, float, str, bool) and use the type

# Integer variable
roll_no = 1
weight = 70.5
name = 'Mahendra'
is_student = True

print(type(roll_no))
print(type(weight))
print(type(name))
print(type(is_student))

<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
```

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In [ ]: # 3. Write a program that takes two numbers as input from the user and prints the sum

num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

sum_result = num1 + num2
difference_result = num1 - num2
product_result = num1 * num2
division_result = (num1 / num2) if num2 != 0 else "Undefined!"

print(f"Sum: {sum_result}")
print(f"Difference: {difference_result}")
print(f"Product: {product_result}")
print(f"Division: {division_result: .2f}")

Sum: 30.0
Difference: 10.0
Product: 200.0
Division: 2.00
```

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In [6]: # 4. Store your full name in a string variable and print its length, uppercase version

full_name = "Mahendra Chaudhary"
```

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print(f"Length of full name: {len(full_name)}")
print(f"Uppercase version: {full_name.upper()}")
print(f"Lowercase version: {full_name.lower()}")
print(f"Full name with underscores: {full_name.replace(' ', '_')}")
```

Length of full name: 18

Uppercase version: MAHENDRA CHAUDHARY

Lowercase version: mahendra chaudhary

Full name with underscores: Mahendra_Chaudhary

In [9]: *# 5. Create a list of 5 favorite movies, print each movie with its index number (star*

```
favorite_movies = ["The Shawshank Redemption", "The Godfather", "The Dark Knight", "P

print("Favorite Movies:")
for index, movie in enumerate(favorite_movies, start=1):
    print(f"{index}. {movie}")

new_movie = input("Enter one more movie to add to the list: ")
favorite_movies.append(new_movie)

print("Updated Favorite Movies List:")
for index, movie in enumerate(favorite_movies, start=1):
    print(f"{index}. {movie}")
```

Favorite Movies:

1. The Shawshank Redemption
2. The Godfather
3. The Dark Knight
4. Pulp Fiction
5. Forrest Gump

Updated Favorite Movies List:

1. The Shawshank Redemption
2. The Godfather
3. The Dark Knight
4. Pulp Fiction
5. Forrest Gump
6. Tare Jameen Par

In [11]: *# 6. Write a program that takes a sentence from the user and counts how many times th*

```
sentence = input("Enter a sentence: ")
e_count = sentence.lower().count('e')
print(sentence)
print(f"The letter 'e' appears {e_count} times in the sentence.")
```

There is a huge mountain that region of Europe.

The letter 'e' appears 6 times in the sentence.